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Efficacy and nontarget effects of broadcast treatments to manage spotted lanternfly (Hemiptera: Fulgoridae) nymphs

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Management to control the spotted lanternfly, *Lycorma delicatula* (White), would ideally achieve managers' goals while limiting impacts on nontarget organisms. In a large-scale field study with 45 plots at least 711 m^2, we tested foliar applications of dinotefuran and 2 formulations of *Beauveria bassiana* (Balsamo) Vuillemin, each applied from the ground and separately by helicopter. Applications targeted early instar nymphs. For both application methods, a single treatment with dinotefuran significantly reduced *L. delicatula* numbers, as measured by catch on sticky bands (91% reduction by air and 84% reduction by ground 19 days after application) and by timed counts (89% reduction by air and 72% reduction by ground 17 days after application). None of the *B. bassiana* treatments significantly reduced *L. delicatula* numbers, even after 3 applications. *Beauveria bassiana* infection in field-collected nymphs ranged from 0.4% to 39.7%, with higher mortality and infection among nymphs collected from ground application plots. *Beauveria bassiana* conidia did not persist for long on foliage which probably contributed to low population reduction. Nontarget effects were not observed among arthropods captured in blue vane flight intercept traps, San Jose Scale pheromone sticky traps or pitfall traps, but power analysis revealed that small reductions of less than 40% may not be detected despite extensive sampling of 48,804 specimens. These results demonstrate that dinotefuran can markedly reduce local abundance of *L. delicatula* with little apparent effect on nontarget insects when applied shortly after hatch, and that aerial applications can match or exceed the effectiveness of applications from the ground.

Key words: aerial spray, *Beauveria bassiana*, dinotefuran, invasive species, *Lycorma delicatula*, power analysis

Introduction

Managers working to limit the negative impacts of invasive insects must consider the effectiveness of management practices along with the potential for those practices to harm nontarget organisms or the environment (Newsom 1967, Nash et al. 2010, Matsuda et al. 2020). The targeted pests' habitat preferences, host plants, feeding strategy, phenology, and vulnerability to chemical and/or biological agents all influence the balance between control measures' effectiveness and unintended negative impacts (Kjaer et al. 1998, Liebhold et al. 2016). The value of a management practice also depends on its cost and the ability of managers to deploy it across the targeted area during the time periods when susceptible life stages are present (Epanchin-Niell et al. 2012, Nishimoto et al. 2021). Together, these considerations help identify management practices that can achieve multiple management goals, including minimizing harm to nontarget organisms.

The spotted lanternfly, *Lycorma delicatula* (Hemiptera: Fulgoridae) (White 1845), is a rapidly spreading, univoltine, invasive phloem-feeder that can kill established cultivated grapevines (Dara et al. 2015). This pest causes nuisance to homeowners by excreting large volumes of honeydew that promotes growth of sooty mold on plants and other surfaces (Lee et al. 2019, Urban 2020). First detected in the United States in Berks County, PA in 2014 (Barringer et al. 2015), *L. delicatula* has spread through natural dispersal and accidental human transport to establish populations in 14 states in the northeastern United States as of January 2023 (NYSIPM, 2023). Ongoing management efforts aim to limit spread and prevent establishment of *L. delicatula* in grape and wine producing regions in the western United States (Wakie et al. 2020, Urban et al. 2021). As this pest's range expands, the number of businesses and transportation hubs affected increases, creating a substantial financial burden on businesses faced with quarantine compliance costs (Harper 2019).

One chemical agent available for managing *L. delicatula* is the neonicotinoid dinotefuran. Dinotefuran caused high mortality in *L. delicatula* nymphs and adults in caged field trials (Leach et al. 2019). It is an appealing chemical treatment option because it is absorbed systemically into plant tissue within 36 h of application (Durkin 2009). Absorption helps to ensure that the insecticide targets insects feeding on treated plants rather than all insects that contact the plant, as can occur with long-residual contact insecticides such as synthetic pyrethroids. In practice, dinotefuran is applied using one of several different methods, including trunk injections, bark sprays, foliar sprays, and soil drenches (Durkin 2009). Beginning in 2015, federal and Pennsylvania State agencies have used dinotefuran as their insecticide of choice in their 'trap tree' control effort, which involves the removal of most of the local *Ailanthus altissima* (Mill.) Swingle, which are among, *L. delicatula's* preferred host plants, and the treatment of a few remaining *A. altisssima* trees with dinotefuran applied as a bark spray postbloom (USDA-APHIS 2021). Sampling of *A. altissima* flowers in the growing season after summer application the previous year revealed very low concentrations of dinotefuran (Elmquist et al. 2023), and no detectable dinotefuran residues were found in nearby honeybees' hive products (Underwood et al. 2019). Concern over the potential for neonicotinoids to harm beneficial insects may limit future use of this insecticide. These risks may be reduced by timing applications to avoid bloom of bee-attractive shrubs and trees (Biddinger and Rajotte 2015), but reluctance to apply these chemicals remains.

The entomopathogenic fungus *Beauveria bassiana* offers an alternative to chemical insecticides and has reduced *L. delicatula* numbers in field trials and under lab conditions (Clifton et al. 2020, Clifton and Hajek 2022). Several strains of this fungus are available in different commercial formulations (Zimmerman 2007, Dara and Arthurs, 2019). *B. bassiana* is a widespread, naturally occurring fungal pathogen with short residual activity and a wide host range spanning at least 15 insect orders (Li 1988). Germination and growth of *B. bassiana* spores depend on environmental conditions including temperature and humidity (Devi et al. 2005). Solar radiation also affects persistence of *B. bassiana* conidia (Ignoffo and Garcia, 1992), though vulnerability to sun damage varies among strains (Morley-Davies et al. 1996). *Beauveria bassiana*'s incubation period varies from days to weeks depending on the host and temperature (Zimmermann 2007). Despite the broad host range of *B. bassiana*, there is little evidence for harmful impacts on beneficial insects, in part because strains may be host-specific and specific environmental conditions required for infection may rarely be met for conidia on these hosts (Hajek and Goettel 2007). This bioinsecticide has been used successfully in efforts to control diverse insect pests including the European corn borer (*Ostrinia nubilalis* Hübner) and the Colorado potato beetle (*Leptinotarsa decemlineata* Say) and has been sprayed aerially over large areas of forest in China for control of the pine caterpillar *Dendrolimus punctatus* Walker (Feng et al. 1994).

The central goal of this study was to evaluate efficacy and nontarget impacts of these control measures (foliar application of dinotefuran and *B. bassiana*) under realistic field conditions for landscape-level control of *L. delicatula* nymphs. We applied insecticides in 2 ways: from the ground (using backpack sprayers for *B. bassiana* and tractor-mounted hydraulic sprayers for dinotefuran) and aerially by helicopter. We also compared 2 commercial formulations of *B. bassiana*: one ready-to use, oil formulation which is claimed to enhance residual activity and reduce the requirement for high humidity for infection (Aprehend), and one oil-based formulation that requires dilution in water prior to application (BoteGHA). We applied treatments to relatively large plots (at least 15 m wide by 46 m long) in woodlot strips, which are similar to the mixed forest and forest edge habitats frequented by *L. delicatula*. Because *B. bassiana* products require incubation and each application was expected to yield a lower effect size than dinotefuran, 3 applications of these products were made, whereas dinotefuran was applied once at the maximum annual label rate.

Methods

Study Site

Insecticide trials were conducted at the U.S. Army Corps of Engineers (USACE) Blue Marsh Lake Recreation Area and Pennsylvania State Game Lands 280 in Berks County, PA, USA (Fig. 1), a site that has had high densities of *L. delicatula* since 2018. The climate in Berks County is temperate: temperature averaged 15.2, 21.6, and 25.6 °C for May, June, and July 2020, respectively. Rainfall totaled 82.80, 82.04, and 157.73 mm in May, June, and July 2020, respectively (NOAA 2023, https://www.ncdc.noaa.gov/cag/county/rankings/PA-011/pcp/202007). This landscape is composed of typical habitat for southeast Pennsylvania's rural areas, with fragmented woodlots and farm fields broken by windbreaks. *Lycorma delicatula* was present, and in some locations highly abundant, throughout the Blue Marsh Lake area in 2020.

Study plots were in soil conservation strips that were established in the 1970s and have been allowed to grow without management. These wooded strips contained many known suitable host plants for *L. delicatula*. The preferred host plant *A. altissima* was present in all plots in varying densities. Plots also frequently contained *Juglans nigra* L., *Acer* spp. L., wild grapevines (*Vitis* spp.), and *Celastrus orbiculatus* Thunb., all of which have been observed to serve as host plants for *L. delicatula* (Liu 2019). The adjacent open fields were either planted with agricultural crops including soy, maize, and wheat, or left unmanaged; unmanaged plots contained a mixture of herbaceous and shrubby plants that were mowed seasonally.

Field Experiment

We laid out 42 woodlot plots, each 46.2 m long and 15.4–30.8 m wide, located at several sites around Blue Marsh Lake (Fig. 1). These plots were randomly assigned to 7 treatments: (i) untreated control, (ii) Aprehend (ready-to-use formulation, Aprehend, ConidioTec, LLC, EPA Registration Number 89186-1) applied aerially, (iii) Aprehend applied from the ground, (iv) BoteGHA (requiring dilution, BoteGHA ES, CERTIS BIO, Manufactured by LAM International, EPA Registration Number 82074-1) applied aerially, (v) BoteGHA applied from the ground, (vi) Safari 20SG (active ingredient dinotefuran) (Safari 20SG, Valent USA, EPA Registration Number 86203-11-59639) with nonionic surfactant LI 700 (LI 700, Loveland Products, Inc., CA Registration Number 3704-50035) applied aerially, and (vii) Safari with surfactant

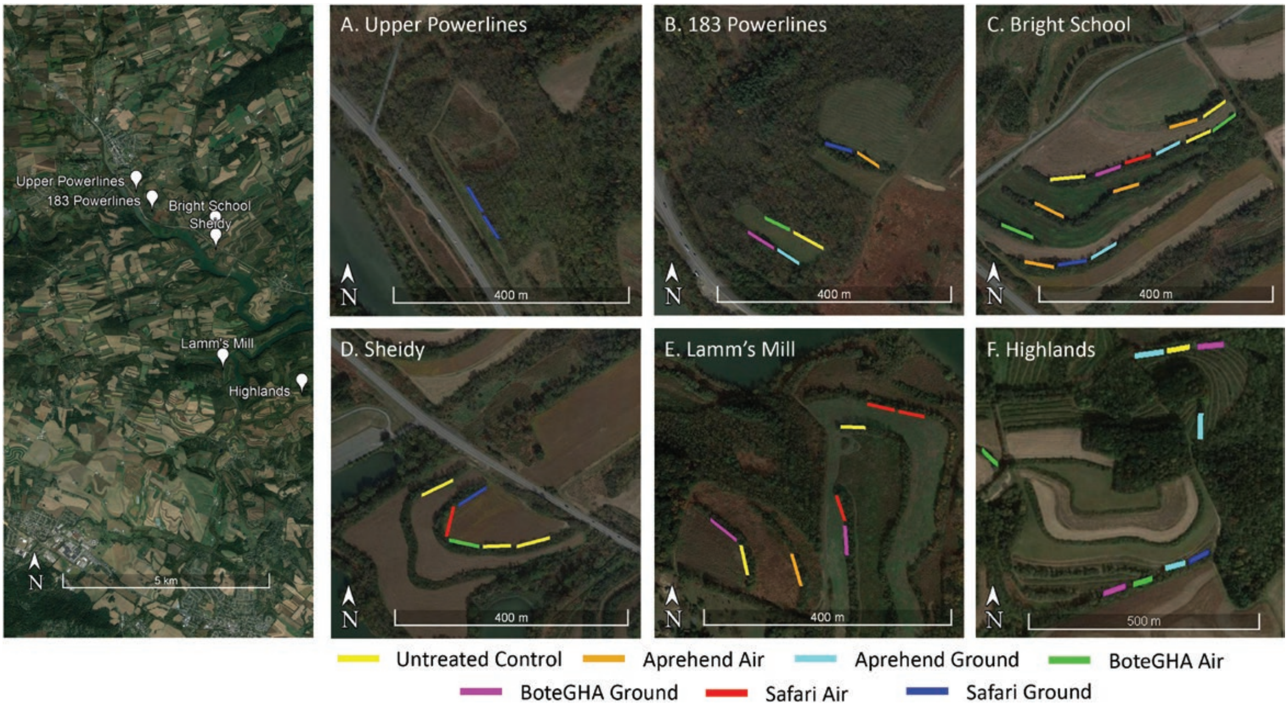

Fig. 1. Experiment layout and treatment assignments at Blue Marsh Recreation Area, Berks County, PA. Images are from Google Earth.

applied from the ground (Table 1). Following environmental review by USDA and USACE, 3 plots were disqualified as treatment sites due to proximity to water. These 3 plots were replaced further from surface water but near original plots to maintain the same types of vegetation and *L. delicatula* densities. The original plots were retained as untreated controls, giving a total of 45 plots. We measured the basal area of *A. altissima* in each plot and estimated the coverage of the known nymphal hosts wild grape and oriental bittersweet.

Application timing was informed by a phenological hatch model for *L. delicatula* (Smyers et al. 2021). The first spray was timed to occur when 90% hatch had occurred, and later sprays proceeded approximately every 2 wk thereafter. Aerial sprays occurred on 15 June 2020 (Aprehend, BoteGHA, and Safari), 29 June 2020 (Aprehend and BoteGHA), and 14 July 2020 (Aprehend and BoteGHA). For aerial applications, we used a Bell 47 helicopter equipped with a spray boom and TeeJet110-degree Acetal-Stainless Steel, flat fan tips (TeeJett Technologies, Glendale Heights, IL, AI11003-VS) fitted to Quick TeeJet adapters with diaphragm check valves (TeeJett Technologies, Glendale Heights, IL, QJ8360-NYB). Aprehend was applied first because the ready-to-use oil formulation cannot be mixed with water. After confirming that spray tanks and lines were dry, we added 56.8 liters of Aprehend and adjusted the nozzle flow rate to ensure a volume application rate of 22.4 liters ha^{-1}. Less than 10 liters of product was required for the application, but excess product was required to fill the tanks and pipes and ensure flow of product to tips. Following Aprehend treatment of assigned plots, tanks and lines were fully drained. Next, we filled the tank with BoteGHA, diluting 1.42 liters of concentrated BoteGHA with 55.4 liters of water. The spray rate was increased to 93.5 liters ha^{-1}. Upon completion of BoteGHA application, tanks were purged over fallow land, far from application sites. Regardless of the volume application rate, applications were calibrated to deliver the label recommended application rate for BoteGHA of 4.94×10^{13} conidia ha^{-1} for both products.

Aerial Safari application occurred only once, on the first date of application. We dissolved 3.06 kg (108 dry oz [US]) of Safari granules in 47.3 liters of water in the tanks. Another 47.3 liters of water was added to tanks with 0.47 liters of LI 700 under constant agitation. The mix was applied at 93.5 liter ha^{-1} to achieve an application rate of 3.0 kg of product (0.6 kg of active ingredient) ha^{-1} with a surfactant rate of 0.5%, volume to volume.

Ground-based sprays of both *B. bassiana* formulations were made on 16 June 2020, 30 June 2020, and 13 July 2020. For the first BoteGHA spray, we used a Solo Model 451 (Solo USA, Inc. Newport News, VA) power backpack mister fitted with Micron Au8000 (Micron Sprayers, Bromyard, UK) nozzle. The first Aprehend application was made with an identical Solo Model 451, but fitted with a Micronex (Micron Sprayers, Bromyard, UK) nozzle. Due to mechanical issues with both Solo sprayers, subsequent applications were made with Stihl Model SR450 (Stihl USA Inc, Virginia Beach, VA) backpack misters, fitted with the same Micron Au8000 and Micronex nozzles. For plots assigned to the Aprehend ground treatment, we used 2% Aprehend concentration and calibrated to deliver 22.4 liters ha^{-1}. In plots assigned to the BoteGHA ground treatment, we applied at a rate of 93.5 liters ha^{-1} of diluted BoteGHA mixture. The dilution was 0.95 liters of BoteGHA concentrate mixed with 36.9 liters of water. We applied 4.94×10^{13} conidia ha^{-1} for both *B. bassiana* products.

Plots assigned to be treated with dinotefuran via ground application were treated once, on 16 June 2020. Plots were sprayed at the rate of 3.0 kg of product (0.6 kg of active ingredient) ha^{-1} with 1.17L LI 700 nonionic surfactant added to 935 liters of water ha^{-1} to achieve a 0.5% volume to volume rate of surfactant. Ground applications of Safari were applied with a John Deer Model 3750 tractor with a 3-point hitch and power take-off drive powering an air blast orchard/vineyard sprayer (model CP14P100P, Durand Wayland, Lagrange, Georgia) with a Gunjet hydraulic handgun attachment (model AA43HC, Spraying System Co., Wheaton, Illinois).

Table 1. Mixing rate and volume application rate for experimental treatments

Treatment	Dates applied	Dilution	Volume application rate	Equipment used
Aprehend Air	15 June, 29 June, 14 July	None	22.45 L/Ha (2.4 gallons/Acre)	Bell 47 helicopter, Tee jet flat fan 110-degree tips
Aprehend Ground	16 June, 30 June, 13 July	None	22.45 L/Ha (2.4 gallons/Acre)	Solo Model 451 back pack mister (first spray), Stihl Model SR450 back pack mister (second and third sprays), Micronex nozzle
BoteGHA Air	15 June, 29 June, 14 July	25 ml/L	93.54 L/Ha (10 gallons/Acre)	Bell 47 helicopter, Tee jet flat fan 110-degree tips
BoteGHA Ground	16 June, 30 June, 13 July	25 ml/L[a]	93.54 L/Ha (10 gallons/Acre)	Solo Model 451 back pack mister (first spray), Stihl Model SR450 back pack mister (second and third sprays), Au8000 nozzle
Safari Air	15 June	3.06 kg Safari/ 94.6 L Water + 0.47L LI-700	93.54 L/Ha (10 gallons/Acre)	Bell 47 helicopter, Tee jet flat fan 110-degree tips
Safari Ground	16 June	3kg Safari/935L Water + 1.17L LI-700	935 L/Ha (100 gallons/Acre)	John Deer Model 3750 tractor, model CP14P100P air blast orchard/vineyard sprayer with model AA43HC Gunjet hydraulic handgun
Untreated Control	NA	NA	NA	NA

[a]An error was made during the second ground-based application of BoteGHA in which 50 ml/L mix rate was applied, resulting in 9.86×10^{13} conidia/ha.

Untreated control plots were not sprayed or otherwise treated in any way.

Assessing Impacts on *L. delicatula*

To measure *L. delicatula* abundance within plots, we installed 2 BugBarrier (Product #BB250 Envirometrics Systems USA, Victor NY) sticky bands per plot. These traps were placed 1–1.5 m high on *A. altissima* trees with diameter at breast height (DBH) 8.5–38.0 cm, with mean DBH 17.5 ± 0.59 cm (±SEM). When a plot did not contain *A. altissima* trees large enough for traps, we installed traps on comparably sized *Prunus serotina* Ehrh. in 2 plots, and on one *P. serotina* and one *J. nigra* in a third plot. We measured DBH of all trees with traps.

Prior to application of treatments, sticky traps were installed in all plots on 2020 8 June. Traps were checked 3 days later on June 11. For each check, we removed and replaced the adhesive film and counted the number of nymphs of each instar stuck to the film. After this initial check, traps were checked on June 14, 23, 26 and July 4, 7, 10, 18, 21, 24, 27, 30. To avoid having personnel in the field during or shortly after sprays, traps were removed prior to spraying, and new traps were installed afterward. Due to these time gaps, the span between check dates was not always 3 days, though traps were always up for 3 days prior to checks.

We also monitored *L. delicatula* abundance using timed counts. To conduct these counts, 4 observers started at separate locations spanning the plot and slowly moved through the plot counting all individuals they could see in 5 min, without moving into areas searched by another observer. This approach followed that used in a prior study on *B. bassiana* effectiveness (Clifton et al. 2020). Timed counts occurred on 2, 9, and 16 July, after nymphs reached the third instar and were large enough to be reliably detected.

Assessing Infection of *L. delicatula* by *B. bassiana*

One day after spraying, approximately 50 *L. delicatula* nymphs were collected from each of the *B. bassiana* treatment and control plots. Field collected nymphs from each plot were placed in pop-up (30.5 cm × 30.5 cm × 30.5 cm) mesh cages (Restcloud, Sichuan, China), and taken to the Penn State Berks campus in Reading, Pennsylvania, where samples from each plot were divided into 3 groups and placed on potted *A. altissima*. Bamboo poles (~1.5 m tall) were used to support mesh nylon insect rearing bags (100 cm × 66 cm) (BugDorm DC3170, MegaView Science Co., Ltd., Taiwan) enclosing foliage. Enclosed plants were randomly distributed in a large shade house. Mortality of nymphs inside each enclosure was recorded daily for 14 days following the first and second spray applications. The same procedure was used following the third spray, but mortality was only recorded 7 and 14 days after collection.

During mortality checks, cadavers were removed from enclosures and transferred to deli cups containing a dry piece of paper towel. Cadavers from each enclosure were kept together until the final mortality check was complete. Cadavers were then checked for mycosis by adding 5 ml of water to a dry paper towel and resealing the cup with the snap-on lid. Cadavers were kept at room temperature in an air-conditioned lab (approximately 20 ± 1 °C) and inspected 14 days later for fungal outgrowth of *B. bassiana*.

Determining Persistence of Conidia on Foliage

In plots that received ground-based applications of *B. bassiana*, leaves were sampled to quantify the number of viable conidia cm^{-2} of leaf immediately after spray applications and 6 and 13 d later. On each sampling date, 3 sets of 10 leaves or leaflets were collected from plots' edges. Samples were taken from a variety of tree species, although most samples were from *A. altissima*. Each set of 10 leaves or leaflets was placed in a 1L zip-lock freezer bag. Because our focus was on quantifying the rate of decline in conidia numbers over time rather than on measuring conidia abundance within plots, samples were not taken from the control or aerial application plots.

Conidia were removed from leaf surfaces by adding 50 ml of sterile 0.1% Tween 80 surfactant (CAS 9005-65-6, Acros Organics, The Hague, Netherlands) to each bag. Bags were agitated vigorously by hand for 30 s and left to soak for 30 min. After soaking, bags were shaken again for 15 s before removing 1 ml of the resulting suspension using a sterile pipette tip. A dilution series using sterile deionized water, up to 10^{-2}, was made for each sample. The surface of 2 replicate 9 cm Petri plates containing sterile selective agar media was inoculated with 0.1 ml from each dilution (0, 10^{-1}, and 10^{-2}) and spread with a sterile spreader. Selective medium was modified from Korosi et al. (2019) comprising: 39 g liter^{-1} of powdered potato dextrose agar (Oxoid, Hampshire, United Kingdom), 0.5 g liter^{-1} of Syllit (Ceres International, West Chester, PA), 0.1 g liter^{-1} of chloramphenicol >98% (CAS 56-75-7, Sigma–Aldrich, Darmstadt, Germany), and 0.5 g liter^{-1} of Streptomycin sulfate, potency 720 µg/mg (Sigma–Aldrich, Darmstadt, Germany).

Plates were incubated at 25°C in the dark for 5 d to allow *B. bassiana* spores to germinate and form small white colonies that were easily distinguished from other microbial growth. Colonies were counted on the 2 replicate plates from the most appropriate dilution (containing 30–300 colonies per plate).

Total leaf area for each set of 10 leaves was measured using the cellphone app "Leafscan" (Anderson and Rosas-Anderson 2017), which typically has error between 2 and 5%, depending on leaf shape and other parameters. Each leaf was removed from the zip-lock bag, blotted dry with paper towels, and placed on a white card within a 15 cm square demarked by 4 corner dots. The cellphone camera was used to photograph each leaf, and the app used the photograph to calculate leaf area. Data from colony counts and the app were used to calculate the number of viable *B. bassiana* conidia per cm^2 of leaf.

Multifaceted Assessment of Nontarget Impacts

To assess impacts on pollinators, a single blue vane trap (Banfield Bio, Inc., Seattle, WA) was installed in each plot. These traps consist of semitransparent cross vanes funneling into a collection jar with ~300 ml of diluted (60:40) ethylene glycol killing/preservative fluid with a few drops of soap to break surface tension. Blue vane traps attract a diverse set of bees, wasps, flies, and other flying insects (Stephen and Rao 2005, Joshi et al. 2015). Blue vane traps were installed 1.22–1.52 m above the ground suspended from branches on plots' outer edges. Traps were first installed on 20 May 2020 and were checked weekly until 12 August. Captured insects were strained from the ethylene glycol and transferred to labeled vials containing 70% EtOH. Specimens were rinsed with soapy water, dried, pinned, labeled, and identified to species. Bees were identified using the Discover Life bee keys at: https://www.discoverlife.org/mp/20q?search=Apoidea#Identification and confirmed voucher specimens. Difficult species were identified by bee specialist Robert Jean at Environmental Solutions and Innovations, Inc. (Cincinnati, OH). Other nontarget specimens were identified to species, when possible, or to family.

Prior insecticide treatments have enabled outbreaks of the widespread pest San Jose scale (SJS, *Quadraspidiotus perniciosus* Comstock, Hempitera: Diaspididae) by eliminating biological control agents (Buzzetti et al. 2015). To assess impacts on this secondary pest and its parasitoid, 1 pheromone baited SJS trap (Trece Pherocon San Jose Scale Traps and pheromone lures, GL/TR-3145-25 and GL/TR-3329-25) was hung in each plot. SJS traps attract SJS males and the biological control agent, *Encarsia perniciosi* Tower (Hymenoptera: Aphelinidae), a parasitoid wasp that uses the pheromone as a kairomone to locate hosts (McClain et al. 1990). SJS traps were installed on 20 May 2020 and checked weekly through 12 August. On each check, the sticky card was collected and replaced. Pheromone lures were replaced every 4 wk. San Jose scales and *E. perniciosi* were identified and counted, while other trapped insects were identified to family when possible and counted.

To assess impacts on ground dwelling arthropods, we deployed pitfall traps in untreated control plots and in all plots treated with dinotefuran. Plots treated with *B. bassiana* did not have pitfall traps installed because prior work showed minimal effects on ground dwelling insects (Parker et al. 1997, Goettel et al. 2021). We installed 3 traps per plot. Traps were 14 cm deep and 11.4 cm in diameter, filled with ~150 ml of 70% EtOH as a killing and preservative agent. Traps were installed with tops level with the soil surface on 17 June 2020 and were checked weekly through 8 July 2020. All captured arthropods were identified to order or family.

Data Analysis

A generalized linear mixed effects model (GLMM) with negative binomial error was fit using the glmmTMB function in R to compare the number of *L. delicatula* on sticky bands (Magnusson et al. 2017, R Core Team 2022). Treatment, DBH of the trap tree, basal area of *A. altissimsa*, *Vitis* coverage, *C. orbiculatus* coverage, date of observation (categorical), and treatment × date interaction were predictors. Location, forest strip ID, plot ID, and tree ID were random effects. Backward selection was based on AIC and carried out using function *buildglmmTMB* (Voeten 2020). To determine which treatments differed from the control on each date of observation, we estimated marginal means and 95% confidence intervals for each date and treatment combination, and compared means, correcting for multiple comparisons within each date using Dunnett's correction. We performed the same model fitting process for timed counts using the sum of the 4 observers' counts as the response.

To assess whether mortality varied among nymphs collected in the field shortly after *B. bassianna* treatments, we considered the proportion of individuals that died after 14 days. We fit a GLMM with beta binomial error distribution and included treatment, spray iteration, and their interaction as predictors, along with a random effect for plot. We selected a beta binomial error distribution after analysis of residuals using the DHARMa package (Hartig 2022) showed overdispersion in a model with binomial error. We estimated the mean proportion of individuals that died for each treatment and spray iteration combination using *emmeans* (Lenth 2022). For post hoc testing, treated plots were compared to the control, and against each other with a Bonferroni correction.

A similar model was used to assess whether the proportion of individuals with mycosis after death varied among treatments. For this analysis, however, 1 treatment combination had no variation in the response (all zeroes), which necessitated the use of Bayesian methods. We fit the model in JAGS using the *rjags* package (Plummer 2022). We ran 3 chains for 50,000 iterations, discarding the first 10,000 iterations for burn-in. We thinned output, taking 1 in 10 samples. Model convergence was assessed based on the Gelman-Rubin statistic (R-hat < 1.1 (Gelman and Hill 2007)). Again, we made 2 sets of comparisons: treated plots against control plots for each spray and also each spray against the others for each treatment.

To evaluate the effect of time since spray on the number of spores present on leaves within plots treated with *B. bassiana* from the ground, we fit a linear mixed effects model with the natural log of the number of CFUs cm^{-2} as the response and treatment (BoteGHA or Aprehend), days since spray, spray iteration, and all interactions as predictors. Again, we included a random effect for plot. We employed the *dredge* function (from the MuMIn package, Bartoń 2022) to fit all models containing subsets of these predictors and selected the model with lowest AICc.

To determine whether the numbers of nontarget insects in study plots were affected by treatments, we considered 8 responses:

1. Number of SJS on SJS traps
2. Number of *Encarsia perniciosi* on SJS traps
3. Number of parasitoid wasps on SJS traps
4. Bee species richness in blue vane traps
5. Number of bees in blue vane traps
6. Total number of insects captured in blue vane traps (excluding *Tiphia vernalis* Rohwer, which was caught in extremely high numbers in some plots)
7. Number of spiders in pitfall traps
8. Number of carabids in pitfall traps

We restricted our analyses to 10 June 2020–5 August 2020 to focus on the period when treatments were most likely to have effects.

For each response, we fit a GLMM with negative binomial error using *glmmTMB* (Magnusson et al 2017, R Core Team 2022). Date of observation (categorical), treatment, and their interaction were predictors, and Trap ID was a random effect. To evaluate which treatments differed from the control on each date, we used *emmeans* to calculate the mean response for each treatment by date combination and compared within dates, correcting for multiple comparisons using Dunnett's correction. For analysis of pitfall trap data, we included the number of traps recovered as an offset, due to the loss of some samples as a result of animal activity in plots.

Because Type II error could misleadingly indicate a lack of impact on nontarget insects when in fact some reduction in numbers may have occurred, we estimated the statistical power of our study design for each nontarget insect response variable. To estimate statistical power, we simulated a data set with response values following seasonal trends as observed in control plots. Then, in simulated treated plots, we reduced the mean response on a given date by a given proportion. We then fit a regression model for the simulated data and determined whether the analysis identified that the control and treated plots differed. This process was repeated for all dates of observation and for a range of proportional reductions. For San Jose Scale catch, we simulated 25%, 50%, 100%, 200%, and 300% increases in numbers, which could potentially indicate outbreaks of this pest. For all other responses, we simulated 90%, 80%, 60%, 40%, and 20% reductions. Replication matched that used in the field (6 treated plots, 10 control plots). We repeated each simulation 300 times and calculated power as the proportion of simulations in which the difference between control and treated plots was correctly identified with $P < 0.05$.

Results

Insecticide Effects on *L. delicatula* Numbers

Immediately after application, sticky traps in plots treated with dinotefuran captured significantly fewer *L. delicatula* than traps in untreated control plots; this pattern was observed for both modes of application: aerial (ratio of 20 June Safari Air mean catch to Control mean catch = 0.18, P = 0.012) and from the ground (ratio of 20 June Safari Ground mean catch to Control mean catch = 0.02, P < 0.001) (Fig. 2). The best supported regression model retained the following predictors: date, treatment, and the interaction between date and treatment (Supplementary Table S1). For plots treated with dinotefuran aerially, *L. delicatula* catch was significantly less than that in control plots up to 36 days after application (ratio of 21 July Safari Air mean catch to Control mean catch = 0.14, P = 0.003). Numbers were suppressed in plots treated with dinotefuran from the ground for 22 days (ratio of 7 July Safari Ground mean catch to Control mean catch = 0.20, P = 0.034).

Timed counts similarly showed significantly fewer *L. delicatula* in plots treated with dinotefuran than in control plots (Fig. 3). This difference occurred for both application methods and all 3 observation dates. The best supported regression model retained treatment, date, and basal area of *A. altissima* as predictors, but not the interaction between treatment and date. Therefore, the expected ratio between *L. delicatula* counts in treated and control plots did not differ between dates (Supplementary Table S2). Plots treated with dinotefuran aerially had significantly lower counts than control plots (ratio = 0.111, P < 0.001), as did plots treated with dinotefuran applied from the ground (ratio = 0.283, P = 0.006).

We observed no difference in numbers of *L. delicatula* caught on sticky bands between control plots and any of the 4 *B. bassiana* treatments (all P > 0.05). Timed counts in *B. bassiana* plots also did not differ from counts in control plots (all P > 0.05).

Assessing Infection of *L. delicatula* by *B. bassiana*

Mortality of field collected nymphs 14 days after spray application did not exceed 60% for any treatment. Following the first spray, control mortality in field collected nymphs reached 26.6% (95% CI [18.9, 36.0]), and a low level of *B. bassiana* infection occurred in the incubated cadavers (Fig. 4). Only Aprehend applied from the ground showed mortality higher than the control. Significantly higher mycosis rates than the control occurred in the Aprehend Air, Aprehend Ground, and BoteGHA Ground treatments.

Mortality of field collected nymphs following the second spray followed a similar pattern, although in this case both ground applications had significantly higher mortality than the control. Only ground applications had significantly higher mycosis than the control (Fig. 4B). Mortality of nymphs from ground applications of both *B. bassiana* products were approximately 41%, but only roughly half of these deaths were confirmed as infection with mycosis.

The pattern of mortality following the third application was different from the first 2 applications (Fig. 4). A high level of mortality was recorded in nymphs collected from Aprehend air plots (51.9%, 95% CI (41.2, 63.4)), but *B. bassiana* infection was only confirmed in 1.5% of these individuals. Mortality of nymphs from the control, Aprehend ground, and BoteGHA ground application plots were similar (18.1, 23.4, and 20.1% respectively), but a greater proportion (10.9 and 11.9%, respectively) showed mycosis in treated plots.

Determining Persistence of Conidia on Foliage

The number of CFU's on foliage declined over time (Fig. 5). The number of CFU's on leaves immediately following ground application was generally higher in BoteGHA plots than Aprehend plots. CFU's declined at a similar rate following each application, and this rate was similar for both products. Model selection did not retain interactions between days post-spray and either treatment or spray iteration, indicating lack of evidence for differing slopes with respect to time since application.

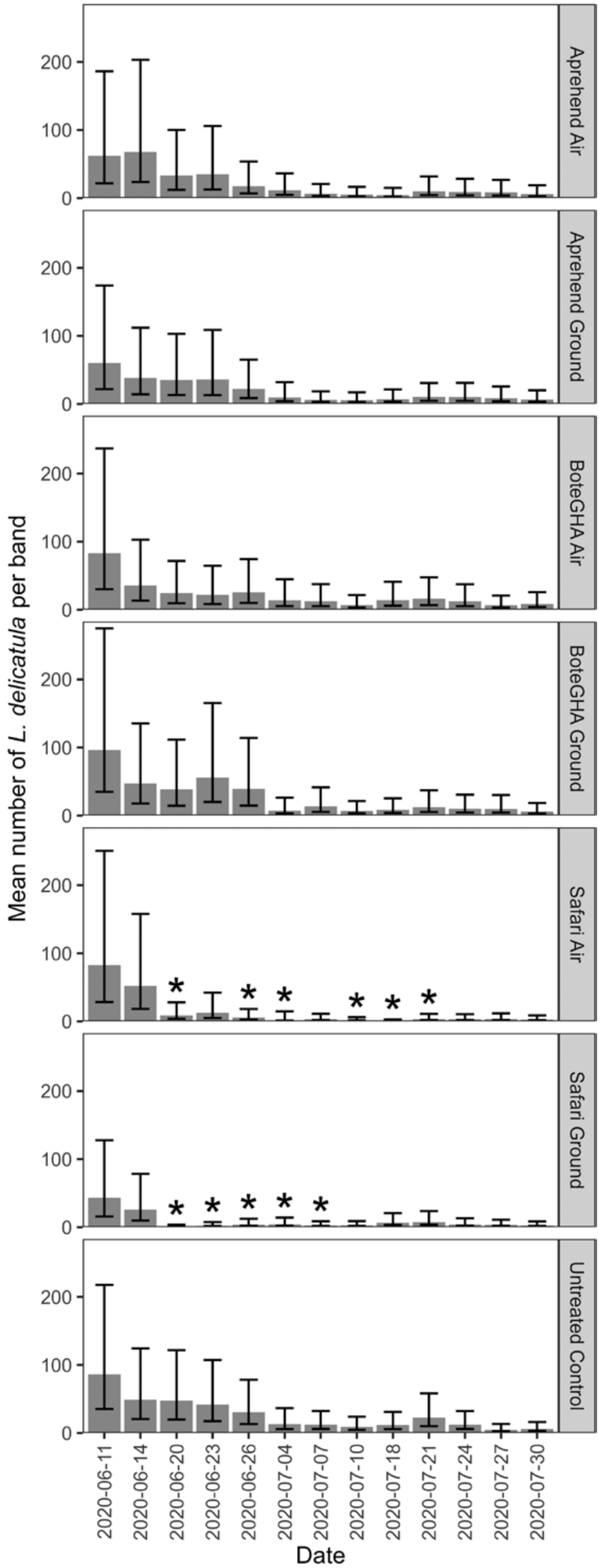

Fig. 2. Both dinotefuran application methods (Safari Air and Safari Ground) reduced numbers of *L. delicatula* caught on sticky bands compared to numbers caught in untreated control plots, as shown by estimated marginal means from a negative binomial generalized linear mixed effects model with 95% confidence intervals for the mean shown over time for each treatment. Bars marked with asterisks were significantly different from the control on that date with α = 0.05.

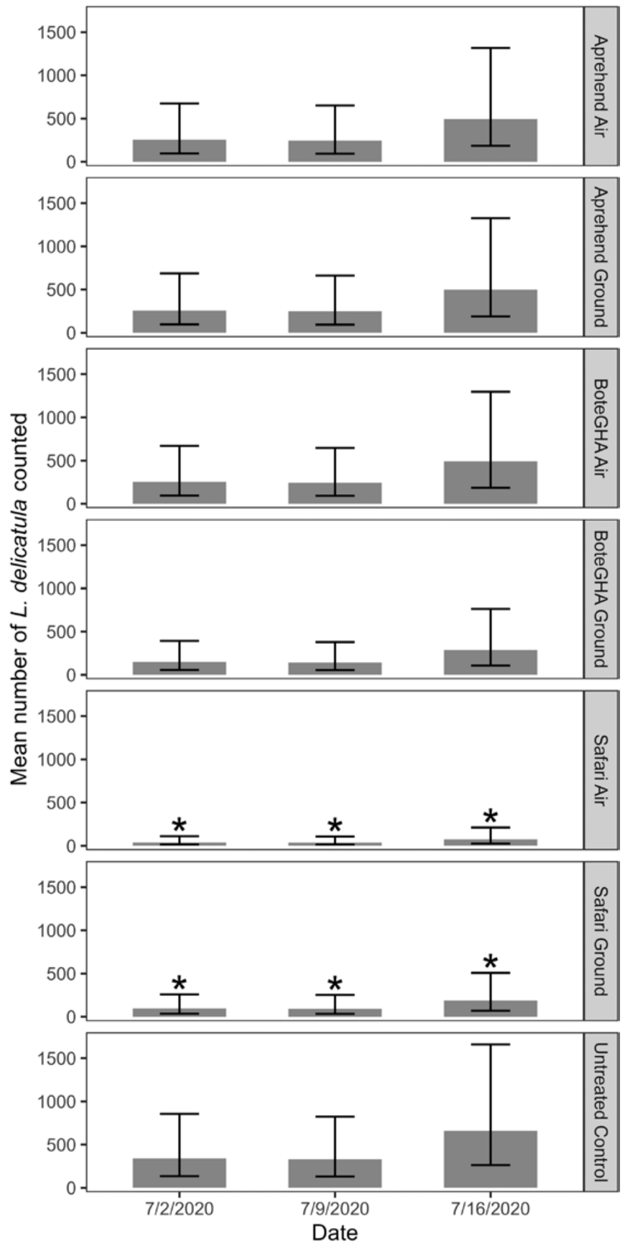

Fig. 3. Dinotefuran reduced *L. delicatula* numbers as measured by timed counts when applied aerially (Safari Air) and from the ground (Safari Ground), as shown by estimated marginal means from a negative binomial generalized linear mixed effects model with 95% confidence intervals for the mean shown over time for each treatment. Bars marked with asterisks were significantly different from the control on that date with α = 0.05.

Insecticide Effects on Nontarget Arthropods

We captured 35,372 arthropod specimens on blue vane traps (mostly bees), 9,145 on SJS traps (SJS, *E. perniciosi* and various families of parasitic Hymenoptera), and 4,287 in pitfall traps. We observed no negative effects of treatments on any of the nontarget insect abundance considered here (Table 2).

Statistical power to detect differences between control plots and plots with simulated reductions in numbers varied among responses and among weeks (Supplementary Fig. S1). For San Jose Scale abundance, increased numbers would indicate a problematic outbreak. We estimate that statistical power exceeded 80% for some weeks' observations with a 3-fold increase in scale numbers, while doubling

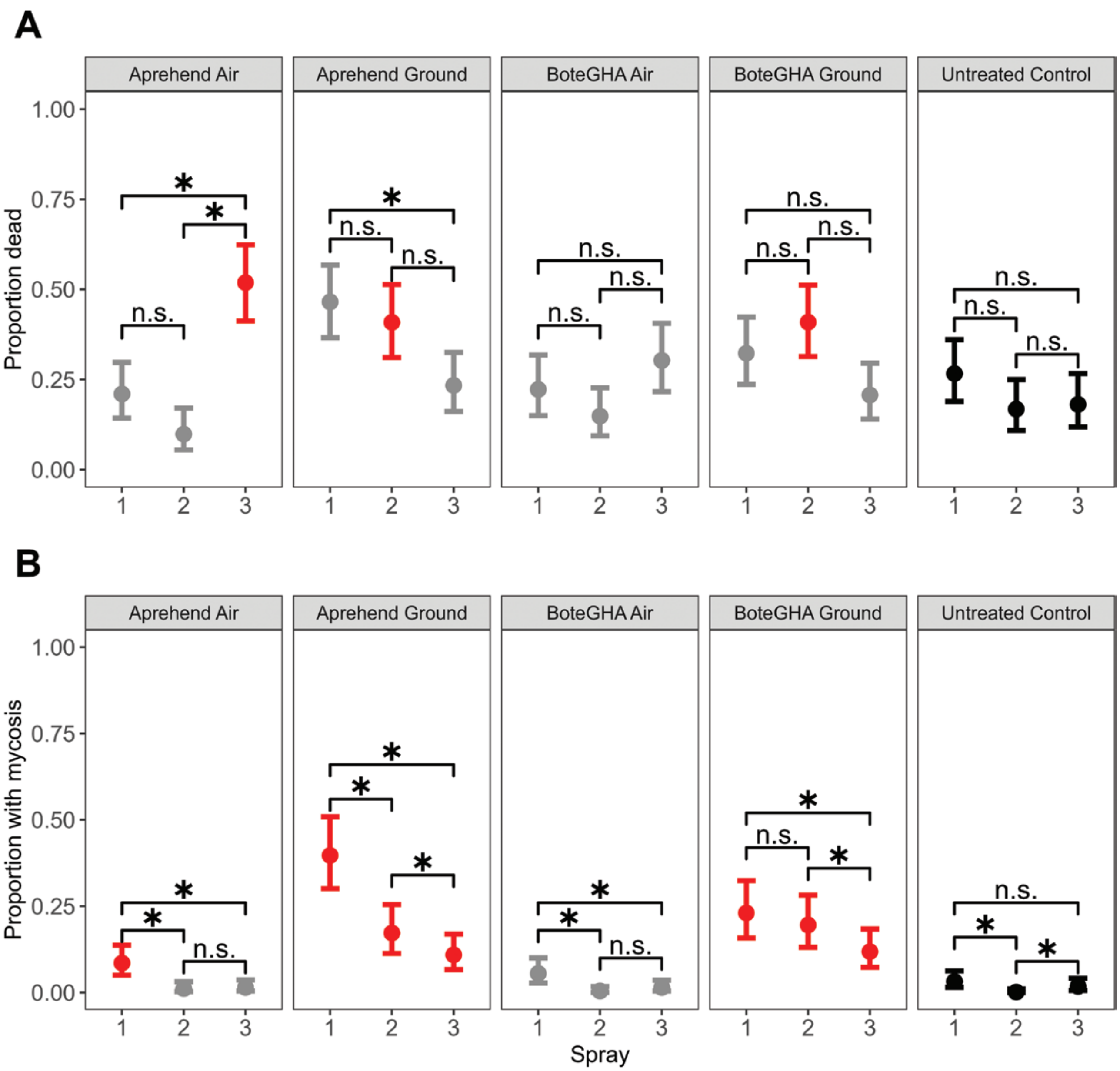

Fig. 4. Mean and 95% confidence intervals for A) mortality of *L. delicatula* nymphs 14 days after collection and B) mycosis of cadavers as a proportion of the total caged population following each of the 3 spray applications. Brackets indicate whether mortality or mycosis varied with each spray for each treatment (n.s. indicates no significant difference between bracketed groups, while asterisks indicate that the bracketed groups were significantly different with $P < 0.05$). Point color indicates whether a treatment was significantly different from the untreated control for that spray iteration. Control points are shown in black, points for treatments that did not differ from the control for that spray are grey, and points where a significant difference was observed are red. Significance was based on a generalized mixed effects model with beta binomial distribution (for mortality) or binomial error distribution (for mycosis).

abundance never resulted in statistical power over 80%. For all other measures of nontarget impact, reductions in the response would indicate a potential harm. All these measures had power over 80% for at least 1 week with a simulated reduction of 80% or more. Some measures exceeded 80% statistical power with a 60% reduction, namely *E. perniciosi* abundance, parasitoid wasp abundance, and bee species richness. No measure exceeded 80% statistical power for simulated reductions of 40% or smaller.

Discussion

Dinotefuran applied from the ground and by air suppressed *L. delicatula* nymph numbers, while 2 formulations of *B. bassiana* did not, despite evidence of infection in nymphs collected from these plots, particularly in plots sprayed from the ground. A single application of dinotefuran reduced trap catch by approximately 99% relative to control plots when applied from the ground, with significant reductions observed for 22 days, and by 82% when applied by helicopter, with significant reductions observed for 36 days. In contrast, 3 applications of *B. bassiana* never resulted in significant reduction of *L. delicatula* in treated plots compared with controls, regardless of application method. Both methods for quantifying *L. delicatula* numbers (sticky bands and timed counts) found the same pattern.

The lack of measurable *L. delicatula* population decline in *B. bassiana* treatments aligned with low mortality and mycosis in field-collected nymphs from these plots. This apparent lack of efficacy

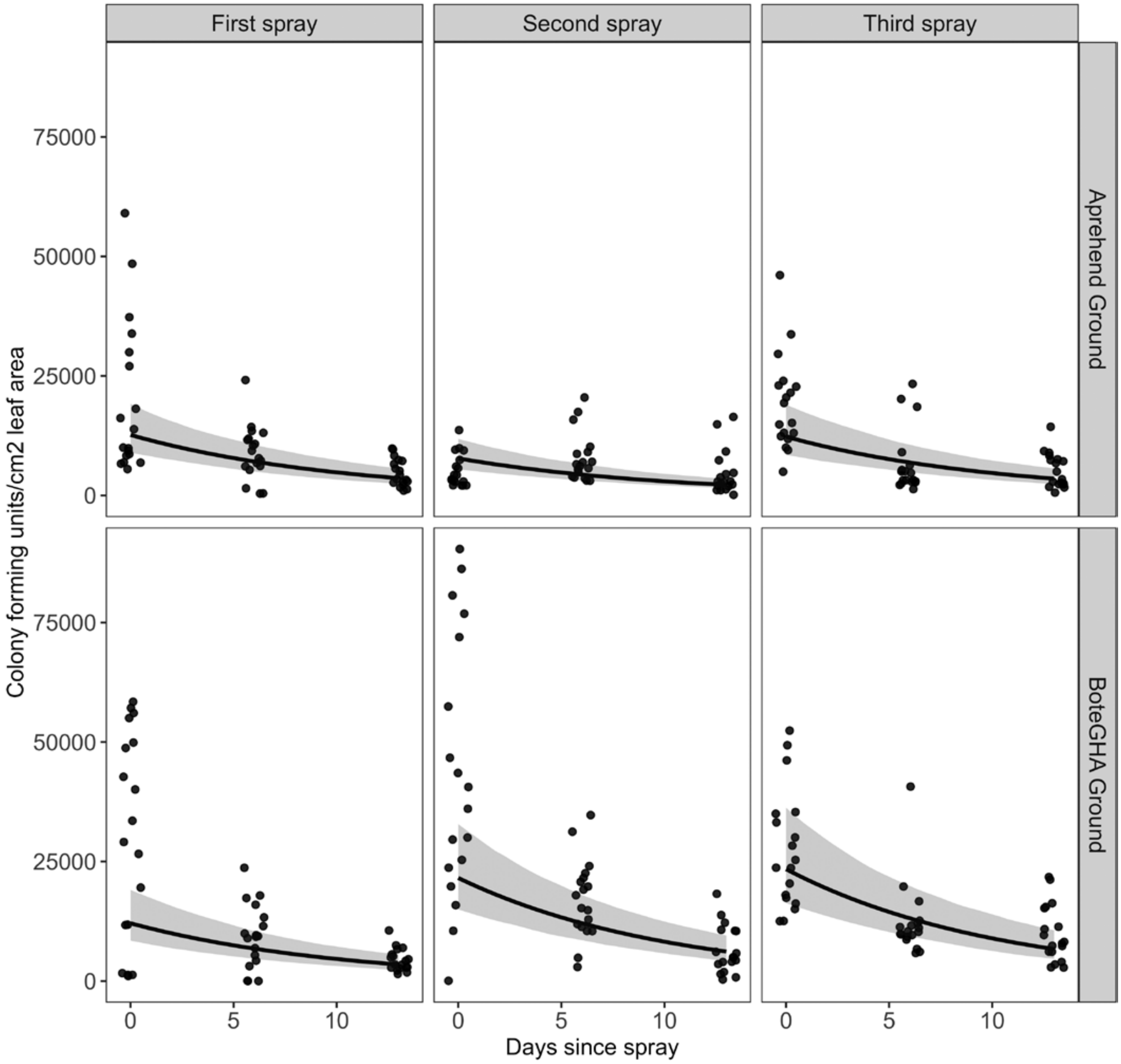

Fig. 5. Number of colony forming units of *B. bassiana* cm^{-2} of leaf area from ground application plots at 0, 6, and 13 days after each of 3 applications. The fitted mixed effects model's estimated mean is shown, with 95% confidence interval indicated by the shaded region.

contrasts with earlier field trials and lab work, which demonstrated that this fungus can cause substantial mortality in *L. delicatula*. Natural outbreaks of *B. bassiana* have killed many *L. delicatula* (Clifton et al. 2019), and, under lab conditions, 90% to 93% of *L. delicatula* nymphs sprayed directly with *B. bassiana* spores died within 14 days (Clifton and Hajek 2022). In a prior field experiment, where 10 m wide vegetation strips were treated with BoteGHA later in the season (targeting fourth instars), researchers documented a 48% reduction in *L. delicatula* numbers relative to water-sprayed controls (Clifton et al. 2020). Differences in the application rate and application methodology may have caused these different outcomes. Dissimilarities in weather following spraying may also partially explain these differences. Weather influences interactions between spongy moth caterpillars and *Entomophaga maimaiga* Humber, Shimazu, Soper, another entomopathogenic fungus, with wetter conditions increasing mortality and warmer conditions reducing mortality (Reilly et al. 2014). Relative humidity (RH) influenced *B. bassiana* replication and conidiogenesis in chinch bugs (Hemiptera: Blissidae) (Ramoska, 1984), with RH of at least 75% required. Relative humidity and temperature also influenced the mortality of *Rhodnius prolixus* Stål (Hemiptera: Reduviidae) treated with *B. bassiana* (Fargues and Luz 2000).

Our work focused on commercially available strains of *B. bassiana* because these bioinsecticides are the tools currently available for managers to use. Other strains may have relatively greater impact on *L. delicatula* mortality and may have differing persistence in the environments where nymphs forage. Other research groups have begun work to identify such strains and develop tools for their mass production, but these efforts require years of work. The efficacy of the tested strains against *L. delicatula* has been confirmed by prior studies (e.g., Clifton and Hajek 2022).

Large scale field studies present challenges that differ from those that experimenters face when working under laboratory conditions, in greenhouses or in field cages. Movement of individuals between plots and from the surrounding landscape cannot be prevented. *Lycorma delicatula* nymphs move tens of meters over the course of just days (Keller et al. 2020), and dispersal to and from plots certainly occurred, as evidenced by increasing numbers in plots treated with dinotefuran after the initial sharp decline (Fig. 2). Natural mortality can also hamper efforts to track responses to management.

Table 2. Estimated ratios between treated plots and control plots for mean values of 8 response variables relating to nontarget impact of treatment, with *P*-value for significance in parentheses

			Week of collection								
Response	Treatment	Trap Source	10 June 2020	17 June 2020	24 June 2020	1 July 2020	8 July 2020	15 July 2020	22 July 2020	29 July 2020	5 August 2020
San Jose Scale	Aprehend Air	SJS trap	NA	NA	NA	NA	0.38 (0.62)	0.48 (0.55)	0.62 (0.80)	0.57 (0.85)	0.49 (0.92)
	Aprehend Ground	SJS trap	NA	NA	NA	NA	0.89 (1.00)	0.48 (0.62)	1.05 (1.00)	2.20 (0.40)	0.63 (0.96)
	BoteGHA Air	SJS trap	NA	NA	NA	NA	0.55 (0.83)	0.89 (1.00)	0.74 (0.92)	2.10 (0.43)	0.32 (0.75)
	BoteGHA Ground	SJS trap	NA	NA	NA	NA	0.48 (0.73)	0.64 (0.85)	0.52 (0.59)	0.45 (0.75)	0.63 (0.96)
	Safari Air	SJS trap	NA	NA	NA	NA	0.84 (0.99)	1.24 (0.96)	1.46 (0.80)	2.45 (0.24)	0.94 (1.00)
	Safari Ground	SJS trap	NA	NA	NA	NA	0.50 (0.8)	0.48 (0.60)	1.59 (0.64)	1.11 (1.00)	0.63 (0.96)
Encarsia perniciosi	Aprehend Air	SJS trap	1.53 (0.85)	NA	0.75 (0.98)	0.54 (0.22)	0.78 (0.85)	0.44 (0.31)	0.79 (0.98)	1.84 (0.68)	1.03 (1.00)
	Aprehend Ground	SJS trap	2.61 (0.17)	NA	1.22 (0.99)	0.79 (0.87)	0.96 (1.00)	0.97 (1.00)	0.84 (0.99)	0.72 (0.97)	1.28 (0.93)
	BoteGHA Air	SJS trap	1.61 (0.80)	NA	1.13 (1.00)	0.62 (0.44)	1.22 (0.91)	0.87 (0.99)	1.31 (0.95)	2.02 (0.55)	0.53 (0.57)
	BoteGHA Ground	SJS trap	2.66 (0.15)	NA	1.79 (0.73)	0.81 (0.9)	1.20 (0.93)	0.75 (0.91)	0.78 (0.97)	1.02 (1.00)	0.45 (0.38)
	Safari Air	SJS trap	2.20 (0.37)	NA	0.60 (0.92)	0.81 (0.9)	1.23 (0.9)	0.61 (0.69)	0.80 (0.98)	1.72 (0.78)	0.99 (1.00)
	Safari Ground	SJS trap	**4.91 (0.00)**	NA	1.06 (1.00)	0.83 (0.93)	1.21 (0.92)	0.67 (0.81)	0.64 (0.87)	0.72 (0.96)	1.14 (0.99)
Parasitiod wasps	Aprehend Air	SJS trap	1.07 (1.00)	0.80 (0.91)	1.35 (0.77)	0.94 (0.99)	0.91 (0.98)	0.79 (0.77)	1.03 (1.00)	1.18 (0.94)	1.39 (0.38)
	Aprehend Ground	SJS trap	1.29 (0.82)	1.02 (1.00)	1.63 (0.33)	1.08 (0.99)	1.12 (0.95)	1.23 (0.80)	1.57 (0.23)	1.49 (0.41)	**1.73 (0.03)**
	BoteGHA Air	SJS trap	1.14 (0.97)	0.97 (1.00)	1.62 (0.34)	0.89 (0.95)	1.23 (0.79)	0.76 (0.69)	1.00 (1.00)	1.04 (1.00)	0.83 (0.85)
	BoteGHA Ground	SJS trap	1.70 (0.18)	1.03 (1.00)	1.34 (0.77)	1.24 (0.71)	1.19 (0.87)	0.94 (0.99)	1.11 (0.98)	0.91 (0.99)	0.88 (0.94)
	Safari Air	SJS trap	1.16 (0.96)	1.09 (0.99)	1.03 (1.00)	0.74 (0.54)	0.97 (1.00)	0.63 (0.26)	0.74 (0.73)	1.15 (0.96)	0.83 (0.85)
	Safari Ground	SJS trap	1.61 (0.28)	0.97 (1.00)	0.95 (1.00)	0.73 (0.49)	0.82 (0.84)	0.61 (0.22)	0.65 (0.48)	0.50 (0.16)	0.68 (0.37)
Bee species richness	Aprehend Air	Blue vane trap	**0.37 (0.03)**	0.67 (0.63)	0.87 (0.98)	0.78 (0.83)	0.75 (0.79)	0.81 (0.90)	0.54 (0.17)	0.72 (0.69)	0.69 (0.65)
	Aprehend Ground	Blue vane trap	0.69 (0.64)	0.89 (0.98)	1.00 (1.00)	1.04 (1.00)	0.94 (1.00)	0.89 (0.98)	0.87 (0.96)	0.80 (0.87)	0.89 (0.98)
	BoteGHA Air	Blue vane trap	0.77 (0.85)	0.94 (1.00)	0.63 (0.52)	1.13 (0.97)	1.04 (1.00)	0.66 (0.51)	0.71 (0.62)	0.84 (0.93)	1.28 (0.83)
	BoteGHA Ground	Blue vane trap	0.71 (0.71)	1.07 (1.00)	1.43 (0.62)	1.48 (0.45)	1.16 (0.96)	0.93 (0.99)	1.08 (0.99)	0.93 (0.99)	1.36 (0.72)
	Safari Air	Blue vane trap	0.57 (0.32)	0.60 (0.43)	1.05 (1.00)	0.66 (0.48)	0.72 (0.69)	0.82 (0.90)	0.88 (0.97)	0.91 (0.99)	0.85 (0.95)
	Safari Ground	Blue vane trap	**0.39 (0.05)**	0.87 (0.97)	0.52 (0.24)	0.61 (0.35)	0.71 (0.67)	0.83 (0.92)	0.79 (0.86)	0.79 (0.86)	0.98 (1.00)
Bee individuals	Aprehend Air	Blue vane trap	0.31 (0.17)	0.64 (0.85)	0.98 (1.00)	0.91 (1.00)	0.68 (0.86)	0.60 (0.68)	0.39 (0.15)	0.59 (0.64)	0.64 (0.76)
	Aprehend Ground	Blue vane trap	0.68 (0.87)	0.98 (1.00)	0.95 (1.00)	1.63 (0.65)	1.73 (0.6)	0.58 (0.62)	0.66 (0.77)	0.65 (0.75)	0.88 (0.99)
	BoteGHA Air	Blue vane trap	0.80 (0.97)	1.29 (0.96)	0.54 (0.63)	1.20 (0.98)	1.67 (0.67)	0.65 (0.78)	0.57 (0.58)	0.62 (0.70)	1.29 (0.94)
	BoteGHA Ground	Blue vane trap	0.59 (0.74)	1.25 (0.97)	1.54 (0.79)	**2.86 (0.04)**	1.75 (0.59)	0.67 (0.82)	0.94 (1.00)	0.99 (1.00)	1.32 (0.92)
	Safari Air	Blue vane trap	0.51 (0.58)	0.64 (0.85)	1.24 (0.97)	0.63 (0.74)	0.65 (0.82)	0.52 (0.48)	0.73 (0.88)	0.82 (0.97)	0.72 (0.90)
	Safari Ground	Blue vane trap	0.45 (0.43)	1.00 (1.00)	0.36 (0.24)	0.57 (0.61)	0.78 (0.96)	0.58 (0.62)	0.83 (0.97)	0.84 (0.98)	1.30 (0.93)
Total insects	Aprehend Air	Blue vane trap	0.46 (0.32)	0.71 (0.86)	1.45 (0.80)	1.22 (0.96)	0.93 (1.00)	0.66 (0.76)	0.44 (0.18)	0.60 (0.58)	0.64 (0.72)
	Aprehend Ground	Blue vane trap	0.73 (0.88)	0.63 (0.70)	0.80 (0.96)	1.49 (0.72)	1.67 (0.56)	0.62 (0.66)	0.70 (0.79)	0.70 (0.79)	0.88 (0.99)
	BoteGHA Air	Blue vane trap	0.70 (0.86)	1.24 (0.96)	0.73 (0.90)	1.15 (0.98)	1.59 (0.66)	0.72 (0.86)	0.63 (0.66)	0.64 (0.68)	1.28 (0.93)
	BoteGHA Ground	Blue vane trap	0.52 (0.48)	0.89 (0.99)	1.56 (0.68)	**2.63 (0.03)**	1.76 (0.48)	0.71 (0.84)	1.00 (1.00)	0.99 (1.00)	1.28 (0.93)
	Safari Air	Blue vane trap	0.50 (0.42)	0.52 (0.45)	1.19 (0.98)	0.72 (0.86)	0.65 (0.78)	0.58 (0.57)	0.74 (0.87)	0.87 (0.98)	0.78 (0.93)
	Safari Ground	Blue vane trap	0.43 (0.27)	1.04 (1.00)	0.49 (0.42)	0.62 (0.67)	0.67 (0.82)	0.67 (0.77)	0.86 (0.98)	0.86 (0.98)	1.25 (0.95)
Spiders	Safari Air	Pitfall trap	NA	NA	0.41 (0.07)	0.69 (0.61)	0.74 (0.73)	NA	NA	NA	NA
	Safari Ground	Pitfall trap	NA	NA	0.85 (0.87)	0.54 (0.33)	0.96 (0.99)	NA	NA	NA	NA
Carabidae	Safari Air	Pitfall trap	NA	NA	1.22 (0.84)	**3.56 (0.03)**	1.72 (0.42)	NA	NA	NA	NA
	Safari Ground	Pitfall trap	NA	NA	0.44 (0.16)	1.76 (0.48)	1.31 (0.8)	NA	NA	NA	NA

Ratios are the estimated mean for plots from treatments divided by the mean in untreated control plots. Values under 1 indicate lower numbers in treated plots, while values over 1 indicate higher numbers in treated plots. Cells with significant differences between the treatment and control (with $P < 0.05$) are in boldface.

In a prior analysis of nymph dispersion patterns based in part on data from the present study, we conducted power analysis to estimate sample sizes needed to reliably detect various reductions in *L. delicatula* numbers at different times throughout the season (Calvin et al. 2021). We found that 16 replicates in a blocked experimental design were required to detect a reduction in numbers of 50% in mid-June. For later dates, a difference that small was not reliably detectable with any of the sample sizes considered. These issues may have contributed to the lack of significant differences between *B. bassiana* treated and control plots.

The observed decline in CFU's from foliage following spray application was similar for Aprehend and BoteGHA formulations, suggesting that the oil formulation did not enhance persistence of infective conidia. CFU counts exhibited a similar pattern of decline to that observed by Clifton et al. (2020). Many publications have demonstrated superior efficacy and persistence of pure oil formulations of entomopathogenic fungi over water-based formulations (Bateman et al. 1993, Jenkins and Thomas 1996, Inglis et al. 2000, Bukhari et al. 2011), but no such enhancement was observed in this trial. The rapid decline in the availability of infective conidia likely contributed to the lack of population control of *L. delicatula* in plots treated with *B. bassiana*.

Given that *L. delicatula* nymphs are difficult to target by direct spray application, effective population control will require insecticide products with good residual efficacy, as demonstrated by this trial with dinotefuran. It is likely that the low levels of *B. bassiana* infection observed in our field collected populations is the result of exposure of those individuals to direct hit by the spray application. This is further supported by the low numbers of infected *L. delicatula* collected from the aerial application plots. Nymphs were collected from all plots 24 h after the spray applications from vegetation 0.5–2 m from the ground. Without access to aerial work platforms, it was not possible to collect individuals from higher in the canopy. This fact may result in sampling bias, particularly in the aerial plots, where direct spray contact would have been more likely to occur higher in the canopy. Assuming nymphs hit by direct spray application in the canopy remained at high elevations, this could explain the difference in infection observed in populations from the ground-based and aerial applications. While our results do not show that aerial application was effective, we cannot rule out its potential future efficacy given the potential sampling bias in this study. However, the requirement for residual efficacy following spray application remains, and evidence presented here suggests that these *B. bassiana* formulations do not have sufficient residual activity to achieve population control under the environmental conditions we encountered. Future work to optimize spray rate and other spraying parameters may yet yield treatments capable of suppressing *L. delicatula*.

In contrast to *B. bassiana* treatments, application of dinotefuran from the ground and from helicopters significantly reduced *L. delicatula* abundance. Foliar sprays tested here are well suited for application of dinotefuran, which is translocated through plants after penetrating leaves (Elbert et al. 2008). The success of aerial applications demonstrates the potential for such methods to be integrated into ongoing management efforts aimed at slowing the spread of *L. delicatula*. Aerial applications may cover large areas quickly and reduce labor hours (Brockerhoff et al. 2010), potentially allowing managers to restrict treatment to seasonal timings when nontarget impacts are limited due to the phenology of beneficial insects. Aerial sprays may also be beneficial in areas where access is limited, such as along railroads, which contribute to dispersal of invasive organisms (Ascensão and Capinha 2017). Future work should evaluate whether similar applications of dinotefuran targeting adult *L. delicatula* can achieve similar levels of control.

While extensive sampling like ours (with 48,804 specimens caught and identified) should be able to document catastrophic local mortality of many nontarget species in response to insecticide application (e.g., Hatfield et al. 2021), we had limited ability to detect small reductions in the abundance of beneficial insects. Reductions in numbers by 40%, for example, had less than 80% statistical power for all measures of nontarget effects. Power analysis is an important component in assessing nontarget impacts, because failure to reject the null hypothesis of no difference between treated and untreated plots may be caused by insufficient sample size in addition to a true lack of differences (Duan et al. 2006). The size and spatial arrangement of study plots may influence assessments of nontarget impacts (Prasifka et al. 2005), with large plots separated by large distances most likely to reflect responses to management (Jepson and Thacker 1990). Our plots were of intermediate size; the same treatments applied to larger plots may have shown greater nontarget impacts because movement from nearby untreated areas into treated plots would be reduced. Also, while our 8 nontarget responses showed no change with any of the treatments applied, species level analyses and community composition analyses are underway and may reveal impacts not described here.

Dinotefuran's short half-life on plant surfaces may explain the apparent lack of nontarget effects. Dinotefuran persists within plant tissues so that only insects feeding on plants are affected. Pyrethroids, which also kill *L. delicatula* (Leach et al. 2019), persist on plant surfaces much longer, and may therefore be expected to have greater potential for nontarget harm, as observed in orchards where the use of neonicotinoids is considered integrated pest management compatible, but applications of pyrethroids can be very disruptive (Biddinger and Hull 1995, Jones et al. 2010). Because neonicotinoids can move into nectar and pollen (Heller et al. 2020), consideration of flowering phenology to avoid bloom of highly attractive bushes and trees should guide application timing for *L. delicatula* control.

Multiple agencies are working to eradicate satellite populations of *L. delicatula* and reduce local abundance and dispersal of individuals. In this study, we highlight the effectiveness of dinotefuran applied in mid-June to reduce local abundance of *L. delicatula*, and we show that spraying using helicopters is a viable approach that may enable managers to quickly cover large areas, including areas that are difficult to reach with ground-based spraying equipment. We were unable to document reductions in *L. delicatula* abundance after repeated spraying with 2 formulations of *B. bassiana*, suggesting that further work is required to understand the environmental conditions and spraying techniques that promote the effectiveness of this bioinsecticide.

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Supplementary Material

Supplementary material is available at *Journal of Economic Entomology* online.

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